

Assignment: Book Data Collection via Web Scraping (Goodreads , Amazon, Bookbub, Kobo, Audible)

1 Objective

In this assignment, you will build a Python web scraping pipeline that, given a book's ISBN-13 number, collects structured data about the book from Goodreads (www.goodreads.com). You will practice HTML/JSON parsing, working with external websites, structured file naming, and organizing scraped data into a clean dataset.

2 Background

Every book can be uniquely identified using its ISBN-13 number (a 13-digit code; older books may only have an ISBN-10, which must first be converted to ISBN-13). Given an ISBN-13, your program must locate the corresponding book page on Goodreads and extract five categories of information, saving each into its own dataset folder using a strict file-naming convention (given below).

3 Tasks

Task 1 — Book Metadata

- Extract the following fields for the given ISBN-13:
 - ISBN-13
 - Author(s)
 - Title
 - Publisher
 - Origin / Country of publication
 - Date of publication
 - Language (the language the book is written in)
 - Genre, (Separated by comma)
- There are five metadata JSON files, one corresponding to each source, with each file containing the metadata collected from its respective source.

Task 2 — Book Coverpage

- Download the book's cover image from Goodreads.
- If multiple cover images/editions are available, save each as a separate numbered file.
- Save into the `book_coverpage/` folder.

Task 3 — Book Blurbs

- Extract the book's description / blurb text as shown on its Goodreads page.
- Save as a plain text file in the `book_blurb/` folder.

Task 4 — Book Reviews

- Collect at least 25 reviews per source for the given book.
- Each review must be saved as an individual text file (not combined into one file).
- Save into the `book_reviews/` folder.

Task 5 — Genres

- Extract the list of genres/tags associated with the book on Goodreads.
- Save as a plain text file (one genre per line) in the `genres/` folder.

4 Folder Structure

Your program must automatically create and populate the following folder structure (do not hardcode paths outside this structure):

```
data/  
  book_metadata/  
  book_coverpage/  
  book_blurb/  
  book_reviews/
```

5 File Naming Convention (mandatory)

Use `<isbn13>` as the ISBN-13 number of the book. All file names must follow the patterns below exactly:

Task	Filename Pattern	Example
Book Metadata	<code><source>_metadata.json</code>	<code>goodreads_metadata.json</code>
Book Cover Page	<code><isbn13>_cp-<source>_<n>.jpg</code>	<code>9780143127550_cp_goodreads_1.jpg</code>
Book Blurb	<code><isbn13>_b-<source>_1.txt</code>	<code>9780143127550_b_goodreads_1.txt</code>
Book Reviews	<code><isbn13>_r-<source>_<n>.txt</code>	<code>9780143127550_r_goodreads_1.txt</code>

Supported sources: [Goodreads](#), [BookBub](#), [Amazon](#), [Kobo](#), [Audible](#).

6 Constraints & Guidelines

- Language: Python 3 (use `requests` / `BeautifulSoup`; Selenium is allowed if the page content is JavaScript-rendered).
- Add a delay of 1–2 seconds between consecutive HTTP requests to avoid overloading the server.
- Your program must accept an ISBN-13 as input (command-line argument or user input) and work for any valid ISBN-13, not just one hardcoded example.

- Handle missing data gracefully (e.g., print a warning instead of crashing if a field/review is not found).
- Do not hardcode scraped values — all data must be extracted programmatically from the live page.